# RAMCO INSTITUTE OF TECHNOLOGY

## Department of Electronics and Communication Engineering

## Academic Year: 2019- 2020 (Even Semester)

## Innovative Practices Description

## UNIT I: INTRODUCTION TO MOS TRANSISTOR

**Degree, Semester& Branch:** VI Semester B.E.ECE A
**Course Code & Title:** EC8095 & VLSI Design
**Name of the Faculty member:** Mr.S.Vijayakumar
**Name of the Topic** Scaling
**Name of the Innovative Practice:** Mind map
**Date & Time:** 16.12.2019& 15 Minutes
**ICT Tool Used:** Black Board

**Description:**

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

***Goals (Learning Outcomes):***

- The students will be able to identify and visually record current understandings.
- The students will be able to summarize key information, clarify relationships or associations between information and ideas and draw conclusions.

***Use of appropriate methods:***

**Justification for choosing the topic using mind map activity:**

MOSFET scaling is the reduction in the parameters (like current, voltage, electric field etc.) due to reduction in length of the transistor with the advancement in technology. Types of Scaling Two types of scaling are common: constant field scaling and constant voltage scaling. Mind map activity helps the students to easily grasp the overview of scaling with the help of graphical representation.

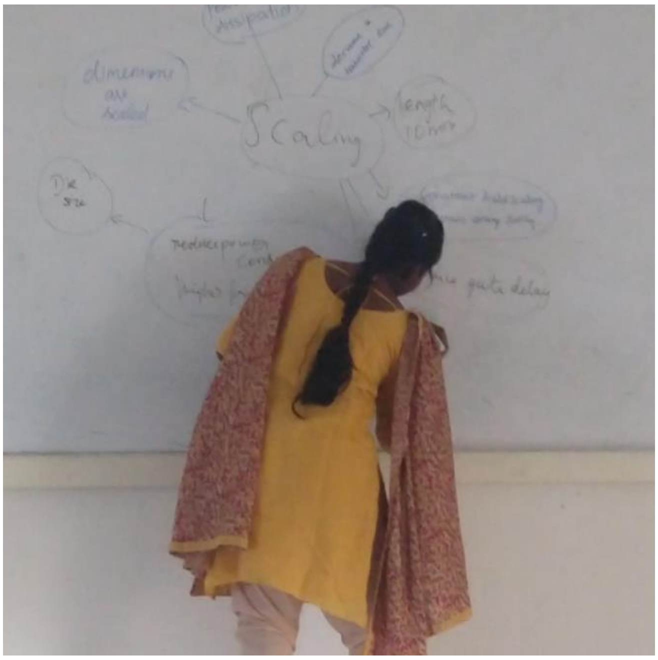
***Effective presentation***

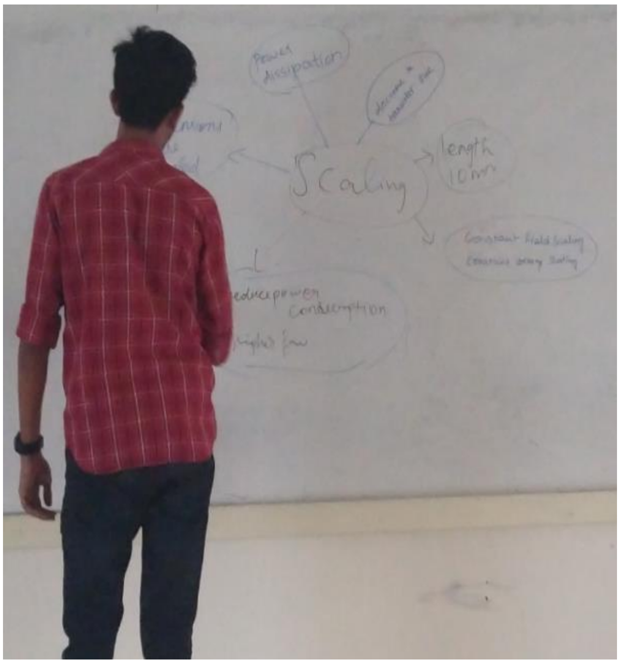
**Details of Implementation:**

I explained to students about the strategy and ensure that students understand that mind maps are personal representations and as such they are not 'right' or 'wrong'. I Select topic scaling of MOSFET and write this in the centre of a blackboard. Students then identify connected key words or phrases related to scaling of MOSFET and write these around the topic, progressively moving to less directly related words.

## Innovative Teaching Method Execution

## SCALING- Mind maps




***Significance of Results:***

**Assessment of Effectiveness/Success of the Activity:**

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

***Reflective Critique:***

**Benefits:**

Mind mapping increases creativity and productivity because it's an excellent tool to let students generate more ideas, identify relationships among the different data and information, and effectively improve students memory and retention. Making a mind map is an excellent way for students to be able to sort through your thoughts and ideas. This activity allows the students to quickly generate creative and even unique ideas in less time.

**Challenges:**

I have planned this activity for 10 minutes but few students not able to recall every details on this topic. So, I gave them another 10 minutes to prepare.

**Changes required for future (if required):Nil**

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 2 | 2 | 2 | - | - | 1 | - | - | - | 1 |

*CO1: The students will be able to realize the concepts of digital building blocks using MOS transistor*

**References:**

**http://www.nhcue.edu.tw/~jinnliu/proj/Device/The%20end%20of%20CMOS%20Scaling.pdf**